MICHAEL WIECK-SOSA

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EDUCATION

Carnegie Mellon University | PhD in Statistics | Advisor: Aaditya Ramdas *May 2027*

- GPA: 3.98/4.00 | Awards: DeGroot-Goel Fellowship (2026), awarded by faculty to one Statistics PhD student per year
- Thesis topics: prediction with transformers, estimation of time series models, and inference for dependence relationships

University of Illinois at Urbana-Champaign | MS in Statistics *May 2022*

- GPA: 3.95/4.00 | Awards: two years of guaranteed assistantships with full tuition waiver and stipend

Fordham University | BS in Mathematics with Minors in Computer Science and Economics *May 2020*

- GPA: 3.77/4.00 | Awards: *magna cum laude* | GRE: 170/170 Quantitative, 163/170 Verbal, 4.5/6.0 Writing

PROGRAMMING LANGUAGES AND TOOLS

- Python expert: extensive experience with NumPy, pandas, Polars, scikit-learn, statsmodels, and PyTorch
- R expert: extensive experience with dplyr, Rcpp, xts, zoo, caret, mgcv, glmnet, parallel, and ggplot2
- SQL, Git, Bash expert: extensive experience with data management, version control, and Linux
- C++ proficient: used for courses in Data Structures and Algorithms
- q/kdb+ proficient: used during internship to work with high-frequency financial data

COURSEWORK

- **Statistics**: Advanced Statistical Theory, Intermediate Statistics, Mathematical Statistics, Advanced Time Series Analysis, Regression Analysis, Computational Statistics
- **Computer Science**: Algorithms, Data Structures, Theory of Computation, Operating Systems, Computer Architecture, Artificial Intelligence, Machine Learning, Data Mining for Listening to the Social Universe
- **Math**: Stochastic Calculus, Measure-theoretic Probability, Functional Analysis, Measure Theory, Interacting Particle Systems, Geometric Flows, Differential Geometry, Lie Groupoids & Lie Algebroids, Topology, Abstract Algebra, Numerical Analysis, Numerical Linear Algebra, Real Analysis, Differential Equations, Linear Algebra, Mathematical Modeling

DOCTORAL RESEARCH

Transformers Meet Invariant Prediction | Aaditya Ramdas

- Developing a method that wraps around pre-trained transformers and selects predictors that are stable across environments

Invariant Prediction for Nonstationary Sequences Beyond the Linear-Gaussian Case | Aaditya Ramdas

- Creating a statistical method that can be used to make machine learning models more robust to nonstationarity

Dynamic Variable Importance for Time Series Foundation Models | Tom Zhang and Chad Schafer

- Quantifying how predictor importance changes across lags and forecast horizons in models like Chronos-2 and TabPFN-TS

Signature Methods for Large Language Models | Leif Weatherby, Tyler Shoemaker, and Cosma Shalizi

- Designing methods for analyzing sequences output by large language models using advanced stochastic process theory

Goodness-of-fit Testing and Confidence Sets through Random Features | Cosma Shalizi

- Building a framework for uncertainty quantification and testing whether the law of a stochastic process is in a model class

Estimating Dynamic Models by Matching Random Features | Cosma Shalizi

- Developing a likelihood-free method for estimating the parameters of complicated models for time series

Conditional Independence Testing for Nonstationary Time Series | Michel F. C. Haddad and Aaditya Ramdas

- Creating a conditional independence test based on regression that is robust to nonstationarity and temporal dependence

GRADUATE RESEARCH ASSISTANTSHIPS

Carnegie Mellon University | NSF Grant | PI: Cosma Shalizi *May 2024-Present*

- Developing theory, methods, and software in Python and R for estimating and inferring the parameters of dynamic models
- Using optimization algorithms to optimize highly non-convex objective functions to fit complicated models for time series
- Applied methods to time series models, state-space models, SDEs, ODEs, and dynamical systems with observational noise

University of Illinois at Urbana-Champaign | FORWARD Data Lab | Computer Science Department *Jan. 2021-May 2021*

- Discovered patterns in cross-platform dynamics on Twitter, Facebook, and Reddit with Hawkes processes using Python

National Center for Supercomputing Applications | Innovative Software and Data Analysis Group *Sept. 2020-May 2022*

- Built confidence bands for trends in concentrations and fluxes of chemicals to measure changes in water quality over time
- Used parallel computing to construct these confidence bands for 1000+ locations in the U.S. and made visualizations using R

INDUSTRY INTERNSHIPS

J.P. Morgan | Quantitative Research | Markets Summer Associate | Equity Derivatives and Macro *June 2023-Aug. 2023*

- Worked with macro traders and quants on a hedging method for derivatives portfolios via multi-period optimization
- Collaborated with energy derivatives traders on improving the statistical techniques used in a new trading strategy

J.P. Morgan | Quantitative Research | Markets Summer Associate | Equity Derivatives *June 2022-Aug. 2022*

- Developed method for adaptively selecting parameters of trade execution algorithms based on real-time data
- Built pipeline for processing market microstructure data using q/kdb+ and making predictions using scikit-learn in Python
- Used Bayesian optimization and walk-forward cross-validation to tune the hyperparameters of machine learning models

RESEARCH INTERNSHIPS

MIT Lincoln Lab | Group 38 *May 2021-July 2021*

- Implemented optical tracking methods for tracking objects in outer space and ran simulations to compare approaches

Corteva Agriscience | Research & Development Division *June 2020-Aug. 2020*

- Analyzed environmental changes based on time series of 50+ weather and soil features from 1,000+ sensor sites in the U.S.
- Used spatiotemporal Gaussian mixture models, dimensionality reduction, and clustering techniques with Python
- Built an interactive Dash application to visualize the statistical analysis of how environments changed over space and time

Fordham University | Robotics and Computer Vision Lab | Computer Science Department *Aug. 2019-March 2020*

- Created a real-time tracking system for the Kihansi spray toad in collaboration with zoologists using deep learning in Python

TEACHING ASSISTANTSHIPS

- For the MS in Computational Finance program at CMU: Simulation Methods for Option Pricing, Financial Time Series, Financial Data Science I and II, and the Machine Learning Capstone Project (x2)
- For the MS in Data Science program at CMU: Time Series and Experimental Design
- For the BS in Statistics and Statistics/Machine Learning programs at CMU: Advanced Data Analysis

PROFESSIONAL SERVICE

- 2026: reviewer of one paper for the Journal of the Royal Statistical Society, Series B (Statistical Methodology)
- 2024-2025: chair of the Statistics PhD student committee at Carnegie Mellon University focused on organizing career events

TALKS AND POSTERS

- 2026: in-person talk on conditional independence testing at the International Workshop in Sequential Methodologies at American University
- 2025: in-person talk on simulation-based inference by matching random features at the Statistical Methods for the Physical Sciences (STAMPS) Research Center's local meeting at Carnegie Mellon University
- 2025: poster on simulation-based inference by matching random features at the Workshop on Neural Simulation-based Inference at Carnegie Mellon University

- 2025: online talk on variable selection for large-scale forecasting for nonstationary time series to researchers at Amazon
- 2025: online talk on conditional independence testing at the Virtual Time Series Seminar's Workshop for Junior Researchers
- 2024: poster on conditional independence testing at the NBER-NSF Time Series Conference at University of Pennsylvania
- 2024: in-person talk on regression for time series at Statistics and Machine Learning Group at Carnegie Mellon University